JORDAN INVESTMENT COMMISSION
الأردن
هيئة الاستثمار

Pre-Feasibility study for the establishment and manufacturing of Bentonite ore

2017

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1. Exclusive summary

Bentonite ores are located in the Ain Al Baida area the reserves estimated of 105 million tons. This material used in many industries, most notably the use of drilling rigs, especially piling works, etc. In order to exploit this ore, the bentonite will be processed and concentrated to produce the required industrial specifications. It will be extracted fromAin al-Baida and transferred to Mafraq area to be manufactured to get the required specifications.

Table1 Project Main Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,467,785
Internal Rate of Return (IRR)	%29.67
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	2,194,055
Job Opportunities provided by the Project	16
Total Revenues (For 10 Years)	21,494,774
Total Operating Expenses (For 10 years)	15,250,542
Total Net Profit (For 10 years)	2,614,660

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

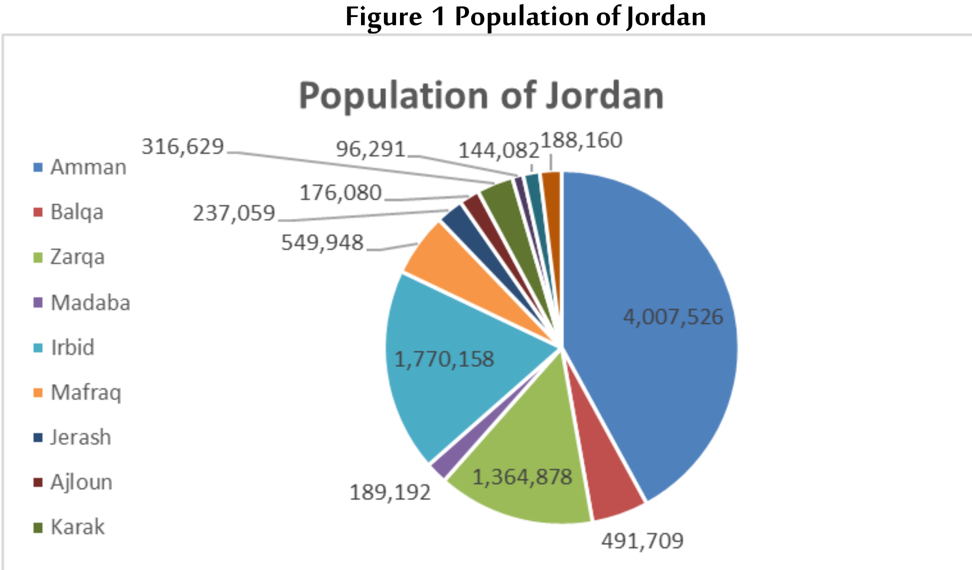
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 1: Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

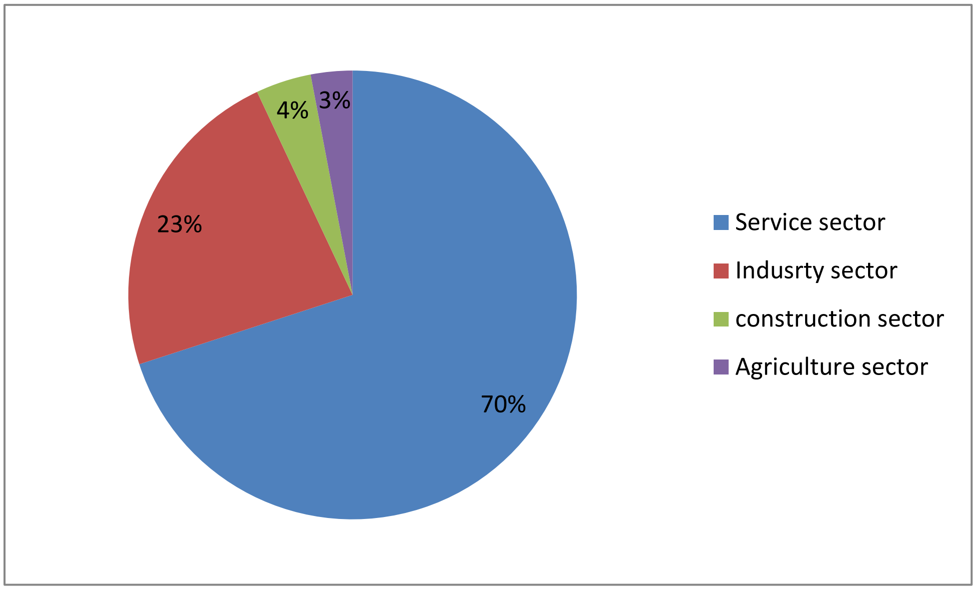
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.
Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.
Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy s lowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.
The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market analysis

Project description

The establishment of a factory with modern processing and concentration technology which includes milling, refining, delineation and hydrocyclonic separation to obtain very soft sizes to be used in different industries, Bentonite is a trade name for a special type of clay consisting mainly of sectite minerals and the predominant mineral is montmorillonite. Bentonite is formed as a result of the transformative processes of volcanic glass which were deposited in shallow lakes in the times of the four era.

Bentonite is a material that is used for the drilling of oil wells and water, but the bentonite should be pumped continuously and regularly to ensure the support of the drilling sides otherwise the bottleneck phenomenon will occured due to unconstant and regular pumping of bentonite during drilling. Its produced in the form of powdered bags (50 kg) available at several factories the most widespread factories are available in India for the Gulf market. It is important to note here that the density of this material depends on the type of soil layers if the soil is weak, the density must be increased and if it is somewhat coherent the density of the material should be decreased.

Chemical properties:

The following table shows the chemical properties of betonite ore in Azraq area.

Table2 Chemical properties of bentonite ore in Azraq area

Compound	Ratio%
Na_2O	% 013
MgO	% 3.47
Al_2O_3	% 20.8
SiO_2	% 55.67
K_2O	% 2.45
CaO	% 2.15
TiO_2	% 2.54
Fe_2O_3	% 13.47

Source: Energy and Minerals Regulatory Commision.

Physical properties:

The following table shows the physical properties of Bentonite:

Table3 Physical properties of bentonite:

property	ratio
Specific weight	2.49-2.72
Surface area	370-487
Ion exchange	53-83
Oils absorption	73-87
Water absorption	115-207
Erosion resistance	80-95
Purification of vegetarian oils	81-99

Source: Energy and Minerals Regulatory Commision.

Distribution of particle size:

The following table shows the percentage of each granular size, as the size of the coarse sand is more than 1,000 microns, the size of the salts is 1000-63 microns and the clay size is less than 2 microns.

Table4 Percentage of particle size.

Partical size	Weight%
m1000+	0.13-0.41
m 63-1000	2.71-4.93
m 2-63	42.14-44.77
m2-	49.89-55.02

Source: Energy and Minerals Regulatory Commision.

Investment Status:

Bentonite ores were explored in 1974. In 1991-1993, the natural resources Authority conducted explorations at the base of Azraq area where wells were drilled and the clay minerals were evaluated The studies indicated that it could be used as a bonding agent in the manufacturing of sand molds for planking and sucking Oils, water absorption, absorption of heavy metals founded in wastewater, as well as its ability to purify edible oils after acid stimulation were evaluated for the first time in 2010

Investment opportunities:

Bentonite deposits are open to investment and mining companies on the basis of detailed exploration, evaluation and exploitation.

Aims of the project

The proposed project aims to:

- The exploitation of Bentonite ores in the Ain Al Baida region, encouraging investors to enter the fields of manufacturing.
- Work to provide employment opportunities that serve the local community and employ Jordanian labor instead of expatriate labor.
- Work to produce materials that serve the local community with the highest possible quality and competitiveness.
- Open horizons for export and marketing of the product globally.

Suggested Product

The use of Bentonite for drilling fluids is the largest use in the world. Its importance in this vital use is through its ability to make a drilling fluid with viscosity and movement properties that makes it able to work drilling craters from deep depths to be removed them to the surface during the drilling process and the ability to keep its suspended in the drilling fluid If the drilling fluid is stopped and rotated inside the well.

- Based on physical and chemical properties, the Jordanian bentonite can be used in the following industries:
- Purification of vegetarian oils
- Wastewater purification
- Water absorption and odors
- Absorption of heavy elements
- The manufacturing of sand molds for molding
- Pelletization

Targeted Categories

• Foundries of Jordan

— Jordan Phosphate Mines Company (Company Foundry)

— Saba Metal Casting Company

— Abdul Raouf Al Ajl & Sons Company

• Water purification plants

— Al Samra Station

— Zi Station

— Shoaib Valley Station

• Cow farms such as:

_Al Bustami Agricultural organisation

_Bittarus Farms in Irbid

_Oil exploration companies

_National Petroleum Company

Market Analysis

The reserve in Ain al-Bayda area is estimated at 105 million tons, while the base of alazraq has not been determined.

Competitors Analysis

Bentonite is one of the raw materials not currently exploited in the Kingdom, so there is no product or expouts.

Price analysis and pricing policy

Bentonite Prices globally:

Bantonite prices ranging between 70-800 JD, based on the specifications of the raw material, its granular size and its industrial uses, noting that the output of the Jordanian raw materials will be divided into three categories:

First type: Used in industrial fields such as drilling fluid, granular size more than 150 microns and the price of 100 dinars per ton.

Second type: Used for the purposes of casting metals, paints and adhesives, granular size 44 microns and the price of 150 dinars per ton.

Third type: Very soft used in medical products granular size less than 18 microns and the price of 200 dinars per ton.

Exploiting these materials will be economically by establishing a factory with a production capacity of 15,000 tons / year at the rate of 300 working days per year, 50 tons / day 8 working hours. Table (5) below shows the expected annual revenues of the project

Table 5 Annual Project Revenues

Bentonite	% Ratio	Annual production (tons)	Price(JD)	Annual Revenues (JD)
First Type	65	9,750	100	975,000
Second Type	20	3,000	150	450,000
Third Type	15	2,250	200	450,000
Total	**100**	**15,000**	**---**	**1,875,000**

Market Share

Due to the absence of any company producing and manufacturing bentonite ores in Jordan, the project will be economically feasible. The local market will be supplied and open for export to the neighboring countries, especially the Gulf countries, because this crude is used in several oil wells. A plant to extract bentonite ores from Ein El Baida area and transfer it to Mafraq to carry out processing and concentrating operations to produce it according to the required industrial specifications. Accordingly, a plant with a production capacity of 15,000 tons / year will be constructed at a rate of 300 working days per year, to 50 tons / day at an average of 8 work hours.

Expected Revenues Analysis

Projected annual revenues for the project

Table 6 Project annual revenues for the years (2018-2028)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
The selling price of 1 ton Bentonait first type	100	103	106	109	113	116	119	123	127	130
The selling price of 1 ton Bentonait second type	150	155	159	164	169	174	179	184	190	196
The selling price of 1 ton Bentonait third type	200	206	212	219	225	232	239	246	253	261
Bentonite first type of production ratio	65%	65%	65%	65%	65%	65%	65%	65%	65%	65%
Bentonite second type of production ratio	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%
Bentonite third type of production ratio	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%
Revenues	**1875000**	**1931250**	**1989188**	**2048863**	**2110329**	**2173639**	**2238848**	**2306013**	**2375194**	**2446450**

3. Technical analysis

This section aims to clarify the technical specifications of the project in addition to estimating and reviewing the most important items of investment and operating costs based on the results of the market study.

The technical analysis includes an explanation and a general description of the aspects of the project and its components, as well as, the calculation of the construction works and operational requirements of the project, specifying the general specifications on which the costs were based, according to the technical team estimations.

Human resorces required

The following table shows the number of employees required for the project:

Table7 Employees and their duties in the factory

Job title	No.
Project manager(mining engineer)	1
Accountant	1
Mining technician	2
Maintenance technician	2
Labour	1
geologist	5
Security gaurd	2
Driver	1
Cleaning labour	1
Total	**16**

The following is a job description of the project's employees:

- **Project Manager:** works independently and within the authority given to him, proposes policies and work plans, the representation of the factory and plan, manage the work of the factory, follow-up implementation of plans and programs in accordance with the policies goals and objectives of the plant to optimize the interests of employers, supervise the department of administrative and financial in comply with managing concerned instructions of subordinates and developing of their skills, the development of procedures and the provision of occupational and public safety and health requirements.

- **Accountant**: Incharge of costs of the plant, in addition to registration of all accounting aspects on the books of accounts.
- **Mining Technician**: Analyzing and studying the mineral resources, and ensures the safe economic operation of mines.
- **Maintenance Technician**: change and repair equipment in all locations according to the plant maintenance plan, Perform preventive maintenance checks and provide all services to other units as required, in addition to implementing the instructions regarding mechanical maintenance procedures.
- **Labuors**: They are the workers who do drilling and extraction work.
- **Geologist**: implementing researches of the ore reserves in order to increase it and exploited by an optimal method.
- **Guard**: Implement the security instructions approved to maintain the security system of the site.
- **Driver**: transfers bentonite ores to the factory for recycling
- Cleaning labour: cleaning and cleaning works to keep the place cleaningof management offices, bathrooms, and all facilities.

Location analysis

The Bentonite is located in two locations, alazraq Base and the Ain Al-Bayda area, about 102 km northeast of the capital, Amman, as shown in Figure 1 below:

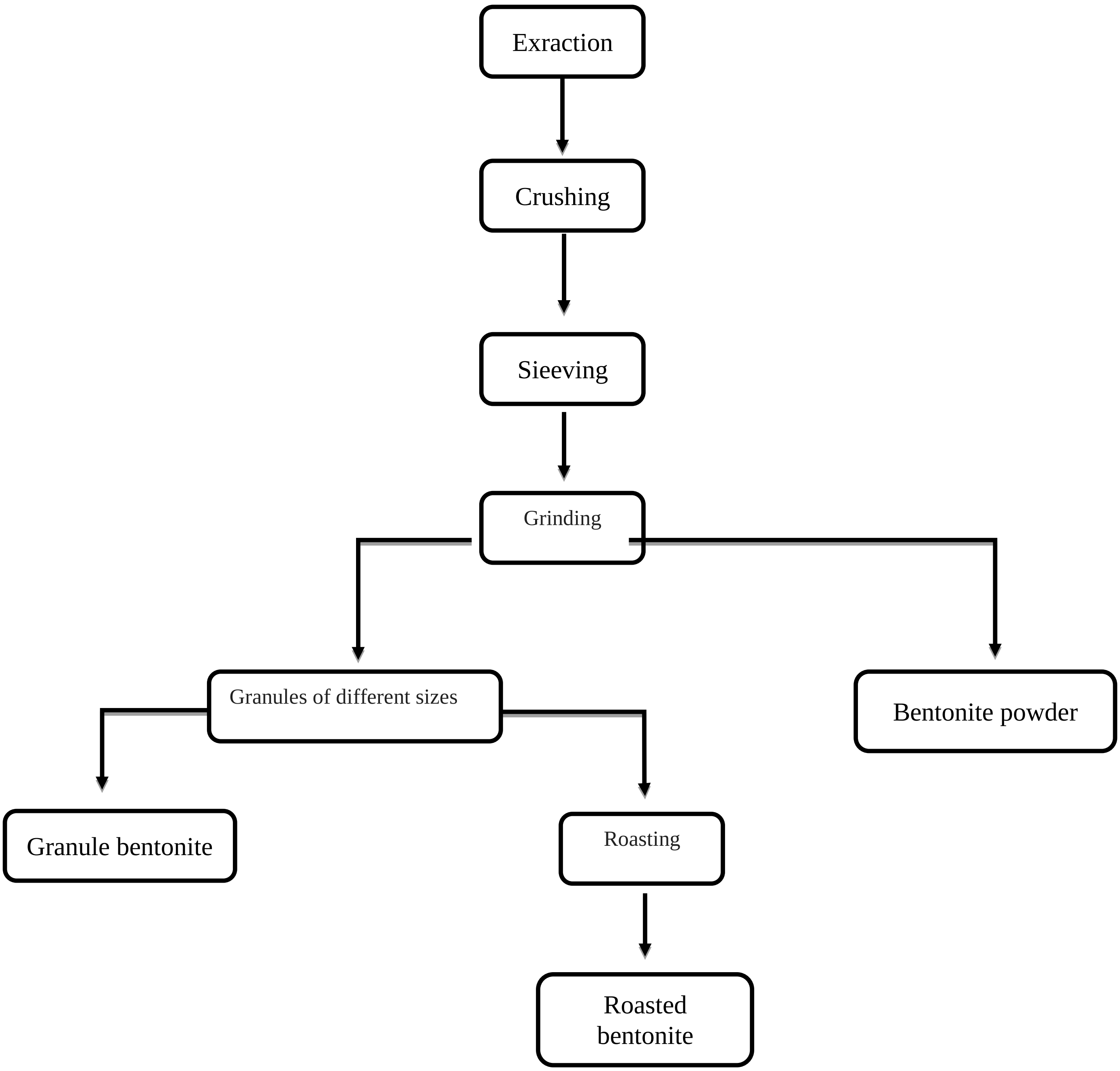
Figure 1: Bentonite's location in Jordan, Source: Energy and Minerals Regulatory commision.

The alazraq base is a closed basin with an area of about 150 km2. Both areas have a slightly topographical elevation (about 510 m) above sea level. Samectite, mixed yacht and cobaltite are the most important minerals found in clay. While quartz, feldspar and calcite are impurities associated with clay minerals, the study areas can be accessed by many ways.

To the estimated reserves of 105 million tons of in the Ain Al Baida area, ore will be extracted from this area and transported to the Mafraq area where the plant will be built due to the availability of manpower, services and infrastructure suitable for the establishment of the plant.

Description of production process and manufacturing steps

After the extraction process carried out, the ore will be of transported the crude to the factory site to stand the process of manufacturing by breaking the large blocks of ore through crusher, followed by the drying process of the ore using the roaster and then to be inserted into the ball mill to obtain the required sizes.



Technical requirement analysis

Land and construction works

The extraction will take place in the area of Ein al-Baida where it was assumed that a 50-dunam plot of land would be leased after conducting the exploration on an area of 1 km2. Table (8) below shows the fees required for the project according to the recommendations of the Central Committee for State Property / Energy Sector Regulatory Commission

Table8 Mining charges. (JD)

The fees	Mining fees (JD)
Fees of prospecting and exploration license of 1km^2	**200**
Fees of annual mining of 1km^2	**500**
Fees of prospecting license granting	**500**
Fees of rent for prospecting aspects about 1km^2	**800**
Fees of annual rent is 25 JD for one dunom, 50 dunom	**1,250**
Total	**3,250**

The raw materials extracted will be transfered from the site to the factory. Ten dunums of land will be purchased in the Mafraq area, where the price of one dunam is about 7,000 Jordanian Dinar, the total cost of the land will be 70,000 Jordanian Dinars.

The following table shows the construction work required for the project:

Table9 Construction works required for the project.

construction works	Area m^2
Building works	200
Construction of Hangar	1000

Equipments and machines

The following table shows the machines and equipment required for the project as follows:

Table 10 Machinery and equipment

Machine	Number
Loader	2
Crushing machine	1
Giinding machine	1
Roasting machine	1
Sieving machine	2

Furnitures

Furniture were estimated based on the needs of the factory, including computers, tables, chairs, carpets, meeting requirements and others.

Requirments and legal procedures

The following figure shows the procedures for obtaining a mining license and mining rights based on the Energy and Metals Regulatory Authority.

Procedures for the exploration license and mining rights

Mining rights

An application for mining rights to be presented attached with approval coordinates by the commission based on the technical report.

Approval of land and survey department or the Jordan Valley Authoristy for treasury lands

The recommendation to the board for mining rights approval will be carried out after presenting the following documents:

- An application request for mining rights.
- Exploration license.
- Aproval of lands and survey department.
- Environmental approval
- Technical report summary.

cabinet approval of the mining right.

Issuance of mining rights after getting a land lease from department of lands and survey, presenting a judicial guarantee and paying the required fees to the commission.

Exploration licenses

Submitting a call request for exploration license attached with of the area coordinates the region and identify the quality of the ore

Provide the following documents:

- registration certifical from the Ministry of Industry and Trade.
- proposed exploration plan.

Site visit by Ministerial committee

Aproval of lands and survey department or on the Jordan Valley Authority for treasury lands

After getting the approvals the recommendation to the board of the commision to grant

The liecense will be issued after presenting a land lease from treasury lands department and paying the fees.

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is %10.75 as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table11 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table12 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table13 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table14 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	60.00%
Utilities Expenses from Total Revenues	2.5%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table15 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	**Value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	20.00%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table16 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	587,114
Loans Value	880,671
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	880,671
Loans Percentage	%60

Equity	
Cost of Equity	%15.5
Equity Value	587,114
Equety Percentage	%40
Gross	
Project Value	1,467,785
Weighted Average Cost of Capital	**%10.75**

Capital Expenditures

The estimated cost of the project is 1,467,785 JD and it covers construction works costs, , machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 17 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	200	200	40,000
Hungers works	100	1,000	100,000
Other Works			14,000
Total			**154,000**

Table 18 machinery and equipments Capital Expenditure

Description	No.	Cost(JOD per	Gross cost
Loader	2	60,000	120,000
Crushing machine	1	40,000	40,000
Giinding machine	1	70,000	70,000
Roasting machine	1	120,000	120,000
Sieving machine	2	25,000	50,000
Total			**400,000**

Table 19 Furniture and furnishings Capital Expenditure

Description	No.	Cost(JOD per	Gross cost
furniture	-	-	120,000
Total			**120,000**

Table 20 vehicles and trasnportations Capital Expenditure

Description	No.	Cost(JOD per	Gross cost
Pick up	2	23,000	46,000
Tipper	2	200,000	400,000
Cost			**400,000**

Table 21 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	**2,500**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table22 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	332,578
General and administrative expenses	3	%25	4,900
Marketing expenses	3	%25	1,172
Indirect Salaries	3	%25	27,200
Total			**365,850**

Table 23 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Construction works	154,000	10%	169,400	%11.5
Machinery &Equipment	400,000	10%	440,000	%30.0
Furniture	12,000	10%	13,200	%0.9
Vehicles	400,000	10%	440,000	%30.0
Pre-Operating Expenses	2,500	10%	2,750	%0.2
Working capital	365,850	10%	402,435	%27.4
Total (JD)	**1,334,350**		**1,467,785**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,467,785) JD, where 60% of the project will be financed through loans (880,671) JD, and the remaining (40%) through (Equity) with (587,114) JD. The following table summarizes the general structure of the required financing:

Table24 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	587,114	%40
Loans	880,671	%60
Total	**1,467,785**	**%100**
Use of Fund:	Falue	Ratio
Capital Expenditures	1,062,600	%72.39
Pre-Operating Expenses	2,750	%0.19
Working Capital	402,435	%27.42
Total	**1,467,785**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table25 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	1,125,000	1,158,750	1,193,513	1,229,318	1,266,197	1,304,183	1,343,309	1,383,608	1,425,116	1,467,870
Utilities from Revenues	46,875	48,281	49,730	51,222	52,758	54,341	55,971	57,650	59,380	61,161
Maintenance from Revenues	18,750	19,313	19,892	20,489	21,103	21,736	22,388	23,060	23,752	24,464
Disposables from Revenues	18,750	19,313	19,892	20,489	21,103	21,736	22,388	23,060	23,752	24,464
Others	120,938	124,566	128,303	132,152	136,116	140,200	144,406	148,738	153,200	157,796
Total	**1,330,313**	**1,370,222**	**1,411,329**	**1,453,668**	**1,497,278**	**1,542,197**	**1,588,463**	**1,636,117**	**1,685,200**	**1,735,756**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 26 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Staionary and Printing	600	618	637	656	675	696	716	738	760	783
Transportaton	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance	7,970	8,209	8,455	8,708	8,970	9,239	9,516	9,801	10,096	10,398
Miscellaneous	1,782	1,835	1,890	1,947	2,006	2,066	2,128	2,192	2,257	2,325
Total	**19,601**	**20,189**	**20,795**	**21,419**	**22,062**	**22,723**	**23,405**	**24,107**	**24,831**	**25,575**

Marketing Expenses

Table 27 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	4,688	4,828	4,973	5,122	5,276	5,434	5,597	5,765	5,938	6,116

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table28 Numbers of employees and their wages expected (JD)

Job title	No.	Monthly salary (JD)	Total monthly salaries	Total annual wages
Project manager(mining engineer)	1	1,500	1,500	24,000
Accountant	1	400	400	6,400
Mining technician	2	350	700	11,200
Maintenance technician	2	300	600	9,600
Labour	1	250	250	4,000
geology	5	380	1,900	30,400
Security gurd	2	300	600	9,600
Driver	1	550	550	8,800
Cleaning man	1	300	300	4,800
Total	**16**		**6,800**	**108,800**

Table 29 Expected Annual Salaries and Wges (2027-2018)

المسمى الوظيفي	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager(mining engineer)	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Accountant	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Mining technician	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Maintenance technician	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Labour	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
geology	30,400	31,646	32,944	34,295	35,701	37,164	38,688	40,274	41,926	43,645
Security gurd	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Driver	8,800	9,161	9,536	9,927	10,334	10,758	11,199	11,658	12,136	12,634
Cleaning man	4,800	4,997	5,202	5,415	5,637	5,868	6,109	6,359	6,620	6,891
Total	**108,800**	**113,261**	**117,904**	**122,739**	**127,771**	**133,009**	**138,463**	**144,140**	**150,050**	**156,202**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	169,400	%5.0	%0.0
Machinaries	440,000	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	440,000	%10.0	%0.0

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,062,600	1,072,060	1,081,729	1,091,612	1,101,715	1,112,042	1,122,601	1,133,395	1,144,433	1,155,718
Total Depreciation Expenses	97,790	98,736	99,703	100,691	101,701	102,734	103,790	104,870	105,973	107,102
Total Accumulated Depreciation	97,790	196,526	296,229	396,920	498,622	601,356	705,146	810,015	915,989	1,023,091
Total Additions	0	9,460	9,669	9,883	10,103	10,328	10,558	10,795	11,037	11,286
Total Net Book Values	964,810	875,534	785,500	694,692	603,093	510,687	417,455	323,380	228,444	132,628

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 30 Loan Details

Loan Value	880,671
Annual Interst Rate	9.48%
Loans Period	5
Loan Starts at year	2017
Annual payment	229,241
Number of payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2017				880,671
2018	229,241	83,488	145,753	734,918
2019	229,241	69,670	159,571	575,348
2020	229,241	54,543	174,698	400,650
2021	229,241	37,982	191,259	209,391
2022	229,241	19,850	209,391	0

Income Statement

Table 31Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,875,000	1,931,250	1,989,188	2,048,863	2,110,329	2,173,639	2,238,848	2,306,013	2,375,194	2,446,450
Gross Operating Revenues	1,875,000	1,931,250	1,989,188	2,048,863	2,110,329	2,173,639	2,238,848	2,306,013	2,375,194	2,446,450
Operating Expenses	1,330,313) (	1,370,222) (	1,411,329) (	1,453,668) (	1,497,278) (	1,542,197) (	1,588,463) (	1,636,117) (	1,685,200) (	1,735,756) (
Gross Operating Profit	544,688	561,028	577,859	595,195	613,051	631,442	650,385	669,897	689,994	710,694
Gross Profit Percentage	*%29*	*%29*	*%29*	*%29*	*%29*	*%29*	*%29*	*%29*	*%29*	*%29*
Salaries and Benefits (Indirect Staff)	(108,800)	(113,261)	(117,904)	(122,739)	(127,771)	(133,009)	(138,463)	(144,140)	(150,050)	(156,202)
General and Administraive Expenses	(19,601)	(20,189)	(20,795)	(21,419)	(22,062)	(22,723)	(23,405)	(24,107)	(24,831)	(25,575)
Markeing Expenses	(4,688)	(4,828)	(4,973)	(5,122)	(5,276)	(5,434)	(5,597)	(5,765)	(5,938)	(6,116)
Pre Operating Expenses	(2,750)									
Gross Indirect Expenses	(135,839)	(138,278)	(143,673)	(149,280)	(155,108)	(161,167)	(167,465)	(174,012)	(180,818)	(187,893)
Income before Interest, Depreciation and Tax	408,849	422,750	434,186	445,915	457,942	470,275	482,920	495,885	509,176	522,800

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Fixed Assets Depreciations	(97,790)	(98,736)	(99,703)	(100,691)	(101,701)	(102,734)	(103,790)	(104,870)	(105,973)	(107,102)
Income before Tax and Interests	311,059	324,014	334,483	345,224	356,241	367,541	379,130	391,015	403,202	415,699
Bank Interests	(83,488)	(69,670)	(54,543)	(37,982)	(19,850)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	227,571	254,343	279,940	307,242	336,391	367,541	379,130	391,015	403,202	415,699
Income Tax	(53,014)	(58,369)	(63,488)	(68,948)	(74,778)	(81,008)	(83,326)	(85,703)	(88,140)	(90,640)
Net Profit	174,557	195,975	216,452	238,294	261,612	286,533	295,804	305,312	315,062	325,059
Net Profit Percentage	*%9*	*%10*	*%11*	*%12*	*%12*	*%13*	*%13*	*%13*	*%13*	*%13*
Compulsory Reserves	(17,456)	(19,597)	(21,645)	(23,829)	(26,161)	(28,653)	(29,580)	(30,531)	(31,506)	(32,506)
Retained Earnings	157,101	333,478	528,286	742,750	978,201	1,236,080	1,502,304	1,777,085	2,060,641	2,353,194

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table32 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	174,557	195,975	216,452	238,294	261,612	286,533	295,804	305,312	315,062	325,059
Bank Interests	83,488	69,670	54,543	37,982	19,850	0	0	0	0	0
Depreciation	97,790	98,736	99,703	100,691	101,701	102,734	103,790	104,870	105,973	107,102
Total Operating Cashflows before Additions to Working Capital	355,834	364,381	370,698	376,967	383,164	389,267	399,594	410,182	421,035	432,161
Inventory (Increase/Decrease)	(93,750)	(2,813)	(2,897)	(2,984)	(3,073)	(3,165)	(3,260)	(3,358)	(3,459)	(3,563)
Accounts Paybales (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Receiviables (Increase/Decrease	34,219	1,027	1,057	1,089	1,122	1,155	1,190	1,226	1,263	1,300
Working Capital (Increase/Decrease)	(59,531)	(1,786)	(1,840)	(1,895)	(1,952)	(2,010)	(2,070)	(2,133)	(2,196)	(2,262)
Net Cashflows from Operating Activities	296,303	362,595	368,859	375,072	381,213	387,257	397,524	408,049	418,839	429,898
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(1,062,600)	(9,460)	(9,669)	(9,883)	(10,103)	(10,328)	(10,558)	(10,795)	(11,037)	(11,286)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Cashflows from Investments الاستثمارية	(1,062,600)	(9,460)	(9,669)	(9,883)	(10,103)	(10,328)	(10,558)	(10,795)	(11,037)	(11,286)
Cashflows from Financing Activities										
Capital	587,114									
Return the loan.	(145,753)	(159,571)	(174,698)	(191,259)	(209,391)	0	0	0	0	0
Bank Interest Rate	(83,488)	(69,670)	(54,543)	(37,982)	(19,850)	(0)	(0)	(0)	(0)	(0)
Loans	880,671	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,238,545	(229,241)	(229,241)	(229,241)	(229,241)	0	0	0	0	0
Net (Increase/Decrease) in Cash	472,248	123,894	129,949	135,948	141,869	376,929	386,966	397,255	407,802	418,613
Cashflows at the Beginning of Period	0	472,248	596,142	726,091	862,039	1,003,908	1,380,837	1,767,803	2,165,057	2,572,859
Cashflows at the End of Period	472,248	596,142	726,091	862,039	1,003,908	1,380,837	1,767,803	2,165,057	2,572,859	2,991,472

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 33 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	472,248	596,142	726,091	862,039	1,003,908	1,380,837	1,767,803	2,165,057	2,572,859	2,991,472
Inventory	93,750	96,563	99,459	102,443	105,516	108,682	111,942	115,301	118,760	122,322
Total current Assets	**565,998**	**692,705**	**825,550**	**964,482**	**1,109,425**	**1,489,519**	**1,879,745**	**2,280,358**	**2,691,619**	**3,113,794**
Non Current Assets										
Fixed Assets (net)	964,810	875,534	785,500	694,692	603,093	510,687	417,455	323,380	228,444	132,628
Total Non Current Assets	**964,810**	**875,534**	**785,500**	**694,692**	**603,093**	**510,687**	**417,455**	**323,380**	**228,444**	**132,628**
Total Assets	**1,530,808**	**1,568,239**	**1,611,050**	**1,659,174**	**1,712,518**	**2,000,206**	**2,297,200**	**2,603,738**	**2,920,062**	**3,246,422**
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	34,219	35,245	36,303	37,392	38,514	39,669	40,859	42,085	43,347	44,648
Remaining amount of Loan	159,571	174,698	191,259	209,391	(0)	(0)	(0)	(0)	(0)	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total current Liabilities	**193,789**	**209,943**	**227,562**	**246,782**	**38,514**	**39,669**	**40,859**	**42,085**	**43,347**	**44,648**
Non Current Liabilities										
Long Terms Loans	575,348	400,650	209,391	0	0	0	0	0	0	0
Total Long Term Liabilities	**575,348**	**400,650**	**209,391**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total liabilities	**769,137**	**610,593**	**436,952**	**246,782**	**38,514**	**39,669**	**40,859**	**42,085**	**43,347**	**44,648**
Owners Equity										
Shareholders Contributions	587,114	587,114	587,114	587,114	587,114	587,114	587,114	587,114	587,114	587,114
Statutory Reseve	17,456	37,053	58,698	82,528	108,689	137,342	166,923	197,454	228,960	261,466
Retained Profits	157,101	333,478	528,286	742,750	978,201	1,236,080	1,502,304	1,777,085	2,060,641	2,353,194
Total Equity	761,671	957,646	1,174,098	1,412,392	1,674,004	1,960,537	2,256,341	2,561,653	2,876,715	3,201,774
Total liabilities and equity	**1,530,808**	**1,568,239**	**1,611,050**	**1,659,174**	**1,712,518**	**2,000,206**	**2,297,200**	**2,603,738**	**2,920,062**	**3,246,422**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Free Net Cash flows Table34 Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(1,467,785)	299,053	353,135	359,190	365,189	371,110	376,929	386,966	397,255	407,802	418,613	6,446,756
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash Flows	**(1,467,785)**	**220,131**	**234,702**	**215,548**	**197,871**	**181,556**	**166,499**	**154,336**	**143,057**	**132,597**	**122,897**	**1,892,647**

Table below (38) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table35 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(1,467,785)	299,053	353,135	359,190	365,189	371,110	376,929	386,966	397,255	407,802	418,613
Project Value Returned	1,467,785	1,168,732	815,597	456,407	91,219	(279,891)	(656,820)	(1,043,786)	(1,441,040)	(1,848,842)	(2,267,455)

Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table36 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	3,210,953
Payback Period	3
Internal Rate of Return	%35.00

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table37 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%29.67	4	10.75%
Revenues declined by 10%	%27.11	4	10.75%
Operating Expenses Increased by 10%	%21.46	6	10.75%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,210,953 Jordan Dinars considering that the project provides 16 Job Opportunitites for the governorate residents